

HIV-1 resistance patterns to integrase inhibitors in antiretroviral-experienced patients with virological failure on raltegravir-containing regimens

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Background: Our aim was to study the *in vivo* viral genetic pathways for resistance to raltegravir, in antiretroviral-experienced patients with virological failure (VF) on raltegravir-containing regimens.

Methods: We set up a prospective study including antiretroviral-experienced patients receiving raltegravir-based regimens. Integrase (IN) genotypic resistance analysis was performed at baseline. IN was also sequenced at follow-up points in the case of VF, i.e. plasma HIV-1 RNA > 400 copies/mL at month 3 and/or > 50 copies/mL at month 6. For phenotyping, the IN region was recombined with an IN-deleted HXB2-based HIV-1 backbone. A titrated amount of IN recombinant viruses was used for antiviral testing against raltegravir and elvitegravir.

Results: Among 51 patients, 11 (21.6%) had VF. Four different patterns of IN mutations were observed: (i) emergence of Q148H/R with secondary mutations ($n=5$ patients); (ii) emergence of N155H, then replaced by a pattern including Y143C/H/R ($n=3$); (iii) selection of S230N ($n=1$); and (iv) no evidence of selection of IN mutations ($n=2$). The median raltegravir and elvitegravir fold changes (FCs) were 244 (154–647) and 793 (339–892), respectively, for the Q148H/R pattern, while the median raltegravir and elvitegravir FCs were 21 (6–52) and 3 (2–3), respectively, with Y143C/H/R. The median plasma raltegravir C_{min} was lower in patients with selection of the N155H mutation followed by Y143C/H/R compared with patients with Q148H/R and with patients without emerging mutations or without VF.

Conclusions: Diverse genetic profiles can be associated with VF on raltegravir-containing regimens, including the dynamics of replacement of mutational profiles. Pharmacokinetic parameters could be involved in this genetic evolution.

Keywords: HIV drug resistance, integrase inhibitors, phenotype, genotype, pharmacokinetics, resistance mutations

Introduction

The selection of HIV-1 variants resistant to antiretroviral drugs is a major factor which can limit the efficacy of antiretroviral therapy (ART).¹ Resistance to all classes of antiretroviral drugs has been characterized. Drug resistance is encoded by mutations in the viral components (reverse transcriptase, protease, *env* glycoproteins) targeted by ART.

HIV-1 integrase inhibitors (INIs) act by blocking the integration of viral double-stranded DNA into the host cell's chromosomal DNA.² Two classes of inhibitors have been described. They interfere either with the 3' processing of the viral DNA long terminal repeats^{3,4} or with the strand transfer of viral DNA into the host genome.⁵ Raltegravir is an integrase strand transfer inhibitor which has shown antiretroviral activity in antiretroviral-naïve⁶ and antiretroviral-experienced patients.^{7,8} In the latter case,

raltegravir combined with an optimized background therapy has led to high response rates in patients with resistance to previous antiretroviral classes.

Resistance to raltegravir has been described *in vitro* and *in vivo*. The most frequent primary raltegravir resistance mutations emerging *in vivo* at virological failure are Q148H/R/K and N155H, and to a lesser extent Y143C/H/R.^{9,10} Numerous other mutations considered as secondary raltegravir resistance mutations have also been described. Elvitegravir is another HIV-1 integrase strand transfer inhibitor in clinical development which has been shown to display cross-resistance to raltegravir.^{11,12}

We studied the genetic pathways of emergence of raltegravir resistance mutations in ART-experienced patients, focusing on the dynamics of specific patterns, on their relationship with virological and pharmacokinetic parameters and on their impact on the phenotypic sensitivity to raltegravir and elvitegravir.

Methods

Study population and design

The patients were selected from the ANRS Co3 Aquitaine Cohort, a prospective hospital-based cohort of HIV-1-infected patients in south-western France. Written informed consent was obtained for all patients. The Aquitaine Cohort has an Institutional Review Board (IRB) approval from the Bordeaux University IRB.

The patients originated from a cohort of 51 antiretroviral-experienced patients treated by raltegravir-containing ART in the setting of an expanded access programme in France.¹³ The dosage of raltegravir was 400 mg twice daily. The co-prescribed antiretroviral drugs were chosen on the basis of a baseline genotypic resistance analysis. The patients were followed-up at months 0, 1, 3, 6, 9 and 12 on raltegravir-containing ART, at which times plasma HIV-1 RNA (CobasTaqman HIV assay, Roche Diagnostics, Basel, Switzerland) and CD4+ T cell counts were measured. Eleven patients who had virological failure (VF; defined as plasma HIV-1 RNA > 400 copies/mL after 3 months on raltegravir and/or > 50 copies/mL after 6 months on raltegravir) were included in this study.

Genotypic resistance analysis

Reverse transcriptase, protease, and *env* gp41 for patients having received enfuvirtide, were submitted for genotypic resistance analysis according to the ANRS consensus procedures available at www.hivfrenchresistance.org. The genotypic sensitivity score (GSS) was calculated as the sum of genotypic sensitivities [according to the ANRS genotype-interpretation algorithm, version 17 (<http://www.hivfrenchresistance.org>); 0, 0.5 or 1 if resistant, partially susceptible or susceptible, respectively] to the drugs co-prescribed with raltegravir. Enfuvirtide prescribed in enfuvirtide-naïve patients was considered as an active drug.

Amplification and sequencing of the HIV-1 integrase gene

The complete integrase gene was PCR-amplified from plasma samples collected at baseline and at the time of VF. Plasma (1 mL) was centrifuged at 19000 g for 1 h at 4°C, and viral RNA was extracted from the pellet using the High Pure Viral RNA Kit (Roche Diagnostics). A 10 µL aliquot of RNA was used for RT-PCR (Titan one-Tube RT-PCR kit, Roche Diagnostics) using primers IN12 and IN13. A nested PCR was then performed by using Ampli Taq Gold DNA Polymerase (Applied Biosystems) with primers IN and BH4.

After purification of the amplified DNA (S400 columns, Pharmacia), the integrase gene was sequenced by using two forward (IN1 and IN4542S) and two reverse (IN4764AS and BH4) primers. The complete PCR protocols, sequencing procedures and primer sequences are described at www.hivfrenchresistance.org. The sequence analysis was processed on a Beckmann CEQ 2000 XL Sequencer and the sequences were aligned with the HXB2 reference sequence. We reported the following 47 mutations present at 31 positions previously associated with *in vitro* or *in vivo* resistance to HIV-1 INIs:¹⁴ H51Y; T66I/A/K; V72I; L74I/A/M; E92Q; T97A; T112I; F121Y; T125K; A128T; E138K/A/D; G140R/C/H; Q146K/P; S147G; Q148K/R/H; V151I; S153Y/A; M154I; N155S/H; K156N; E157Q; K160D/N; G163R/K; V165I; V201I; I203M; T206S; S230N/R; V249I; R263K; and C280Y.

Determination of plasma raltegravir concentrations

Blood samples were drawn to determine plasma raltegravir concentrations at the pharmacokinetic steady state 1 month after starting raltegravir as well as at months 3 and 6. Minimum (C_{min}) serum drug concentrations, corresponding to ~12 h after raltegravir ingestion, were measured using a validated HPLC with mass spectrometry detection (LC-MS).¹⁵

Phenotypic sensitivity to raltegravir and elvitegravir

Viral RNA extraction

Viral RNA was isolated, starting either from 256 µL of plasma using the automated QIAamp Virus BioRobot MDx extraction platform (Qiagen), or from 600 µL of plasma using the EasyMAG procedure (BioMérieux) following the manufacturers' instructions.

Amplification of the HIV-1 integrase gene

Starting from viral RNA, cDNA was generated using the Accuscript High Fidelity Reverse Transcriptase (Stratagene) with random hexamers. cDNA synthesis consisted of three steps: 10 min at 25°C; 60 min at 42°C; and 15 min at 70°C. Subsequently, the integrase gene was amplified by nested PCR using forward primers 5'INoutR1 (positions 4059–4081 in HXB2) and 5'INinF1 (positions 4143–4164 in HXB2) and reverse primers 3'INoutR2 (positions 5241–5262 in HXB2) and 3'INinR1 (positions 5195–5217 in HXB2). Both the outer and inner PCRs were performed using the Phusion High-Fidelity PCR Master Mix (Finnzymes). Thermal cycling of both PCRs consisted of a denaturation step at 98°C for 30 s, 30 cycles of 10 s at 98°C, 30 s at 58°C and 30 s at 72°C, and a final elongation for 10 min at 72°C.

Production of replication-competent recombinant viruses

An HXB2-based HIV backbone in which the integrase region was deleted (pHXB2-ΔIN) was used, essentially as previously described.¹⁶ Integrase amplicons were then recombined intracellularly with the pHXB2-ΔIN backbone by Amaxa nucleofection (Amaxa Biosystems) following the manufacturer's recommendations. The cytopathic effect (CPE) was monitored during the course of infection. When full CPE was reached, recombinant viruses were harvested by centrifugation.

Drug susceptibility testing of recombinant viruses

Replication-competent recombinant viruses were titrated and subjected to antiviral testing in MT4-LTR-eGFP cells using raltegravir and elvitegravir at concentrations from 0.1 nM to 3.6 µM and from 0.04 nM to 0.7 µM, respectively. After 3 days of incubation at 37°C and 5% CO₂, infection

was quantified by means of UV microscopy measuring the HIV Tat-induced eGFP expression. Using the IIIB HIV-1 wild-type virus as a reference, fold change (FC) values were calculated.

Results

Baseline characteristics and follow-up

The baseline characteristics of the 11 patients with VF on raltegravir, and the co-prescribed drugs within the optimized regimen, are described in Table 1. The median baseline viral load was 5.4 log₁₀ copies/mL (range: 2.5–6.7) and the median CD4 cell count was 36 cells/mm³ (range: 7–300). The median GSS was 1 (range: 0–3). All HIV-1 strains belonged to HIV-1 subtype B, except for patient 4 who was infected with an HIV-1 subtype G strain.

The evolution of plasma HIV-1 RNA on raltegravir in the 11 patients is shown in Figure 1. Three patients (patients 1, 2 and 11) had an initial viral load drop followed by a rebound to the baseline level; five patients had no (3, 4 and 5) or only a limited (6, 7 and 8) decrease in viral load; two patients (9 and 10) had a suboptimal inhibition of viral replication at month 3 or month 6. The median decrease in CD4+ T cell count during a 1 year follow-up on raltegravir was –9 cells/mm³ (range: –179 to +359).

Selection of INI resistance mutations

The selection of INI resistance mutations by a raltegravir-containing regimen is shown in Figure 1 and in more detail in Table 2. The integrase sequences were determined at baseline (D0) before prescription of raltegravir. Several polymorphic mutations previously related to increased levels of resistance to INI in the presence of primary mutations were present at day 0; V72I was present in seven patients, V20I in five patients, T206S in two patients, V151I in two patients and I203M in one patient. The median number of baseline mutations was 1 (range: 0–3).

Four different genotypic patterns could be characterized when we studied the selection of INI resistance mutations on raltegravir-containing regimens: (i) in five patients, selection of the Q148H/R mutations with co-selection or accumulation of other mutations (G140S/A in five patients, L74M in two patients, and E138A and K156N in one patient); (ii) in three patients, selection of the N155H mutation followed by its disappearance and replacement by a pattern comprising the Y143H/R/C mutations with other mutations (T97A in three patients, L74M in two patients, and G163R and S230R in one patient; raltegravir was specifically stopped between months 6 and 12 in patient 6, with disappearance of the selected mutations); (iii) in one patient with suboptimal response the mutation S230N was shown to have been selected at month 3; and (iv) in two patients no selection of integrase mutations could be evidenced (patients 10 and 11).

Using this limited dataset, no association could be found between the presence of baseline integrase polymorphisms and the selection of either the Q148H/R pathway or the N155H followed by Y143C/H/R pathways.

Plasma raltegravir concentrations

The median raltegravir C_{min} was 250 ng/mL [interquartile range (IQR): 150–350], 300 ng/mL (IQR: 200–350) and 290 ng/mL

Table 1. Baseline and treatment characteristics of 11 patients failing on raltegravir-containing regimens

Patient	VL	CD4	Co-prescribed drugs	RT mutations	PR mutations	GSS
1	5.6	73	3TC/ETR/ENF	41L 44D 74V 103H 181C 184V 210W 190A 215Y	10F 13V 20T 33V 36I 46L 62V 63P 71V 73S 71V 82A 90M	0
2	4.6	16	TDF/FTC	41L 67N 69N 70R 74V 215F 219Q	10V 33F 46I 54V 63P 71V 73S 82A/T 84V 89V 90M	0
3	2.5	254	3TC/TDF/ETR	41 L67N 74V 181C 184V 210W 215Y	10I 20R 32I 33F 36I 46I 53L 54L 71V 82F 90M	0
4	4.7	7	3TC/ddI/ETR	41L 65R 67N 103H 151M 181C 190A 215I/V 219E	10I 32I 33F 36L 46I 54V 58E 63P 69K 71V 76V 82A 84V 89I 90M	1
5	5.2	89	FTC/ETR/DRV/r	41L 44D 67N 69D 70R 74I 100I 103N 184V 188H 215Y 219Q	10I 20R 32I 33F 46I 47V 50V 62V 63P 71V 90M	1
6	6.7	16	ETR/DR/RTV	41L 44D 67N 69D/N 74I 75T 101E 179F 181C 184V 190S 210W 215Y	10F 15V 20R 32I 33F 36I 50V 54A 62V 63P 71V 84V 85V 89V 90M	0
7	6.4	36	ETR/DRV/r/ENF	41L 44D 67N 74V 103N 184V 210W 215Y	10FV 15V 20R 33F 46I 62V 63P 71T 73S 76V 84V 89V 90M	1
8	5.6	22	TDF/FTC/ETR/DRV/r	41L 44D 67N 70R 181C 184V 190A 210W 215Y	10V 15V 20R 32I 33F 46I 54V 58E 63P 82A 84V 90M	0
9	5.4	13	ddI/TDF/ETR/DRV/r	74I/V 106M 115F 184V 190A	10F 20R 32I 33I 36I 54M 71V 73S 89V 90M	2.5
10	3.7	300	3TC/ETR/DRV/r	98S 101E	10F 11I 20I 32I 33I 36I 46I 54V 63P 71V 73S 84V 90M	2
11	5.8	124	FTC/TDF/DRV/r	—	13V 20I 36I 63P	3

VL, viral load (plasma HIV-1 RNA), log₁₀ copies/mL; CD4, CD4+ cell count (cells/mm³); GSS, genotypic sensitivity score [sum of the genotypic sensitivities [according to the ANRS genotype-interpretation algorithm, version 17 (<http://www.hivfrance.chresistance.org>); 0, 0.5 or 1 if resistant, partially susceptible or susceptible, respectively] to the drugs co-prescribed with raltegravir]; 3TC, lamivudine; ETR, etravirine; ddI, didanosine; DRV/r, darunavir boosted with ritonavir; ENF, enfuvirtide; FTC, emtricitabine.

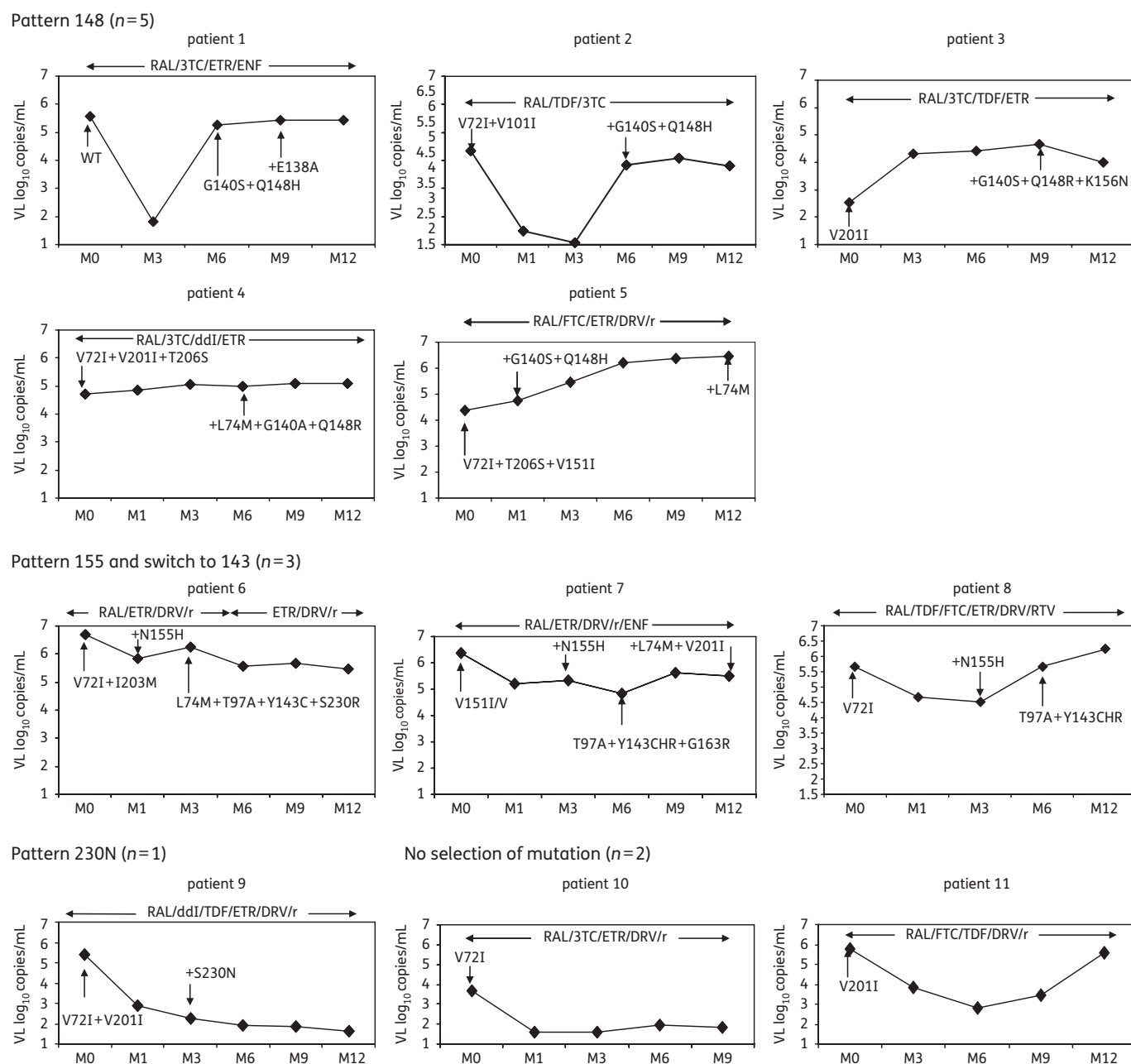


Figure 1. Virological evolution and different patterns of integrase mutations in 11 patients with virological failure on raltegravir-containing therapy. Time of follow-up is expressed in months (M0–M12). Baseline integrase polymorphisms appear at month 0. Additional mutations selected during follow-up are preceded by (+). In patients 6, 7 and 8 the mutation N155H is then replaced by a different pattern. VL, viral load (plasma HIV-1 RNA); RAL, raltegravir; 3TC, lamivudine; TDF, tenofovir diphosphofumarate; ETR, etravirine; ddI, didanosine; DRV/r, darunavir boosted with ritonavir; ENF, enfuvirtide; FTC, emtricitabine.

(IQR: 150–350) at month 1, 3 and 6, respectively. The median plasma raltegravir C_{min} was lower in patients with the selection of the N155H mutation followed by Y143C/H/R compared with the patients with the Q148H/R mutation and with the patients without emerging mutations or with virological response; this difference was statistically significant at month 3 and 6 (Table 3).

Phenotypic sensitivity to raltegravir and elvitegravir

The FC values to raltegravir and elvitegravir are shown in Table 2. The median baseline raltegravir FC was 0.91 (range: 0.69–2.18) and the median baseline elvitegravir FC was 0.94 (range: 0.6–2.12). In samples with the Q148H/R mutations, the median raltegravir FC was 326.48 (range: 154.46–908.45) and the

Table 2. Genotypic evolution of integrase and phenotypic sensitivity to raltegravir and elvitegravir in 11 patients failing on raltegravir-containing regimens

Time on RAL	ART	Viral load (log ₁₀ copies/mL)	CD4 (cells/mm ³)	Integrase mutations	RAL FC	EVG FC
Patient 1						
D0	RAL/3TC/ETR/ENF	5.6	73	—	0.91	0.60
M3	RAL/3TC/ETR/ENF	1.8	78	ND	ND	ND
M6	RAL/3TC/ETR/ENF	5.2	144	G140S Q148H	ND	ND
M9	RAL/3TC/ETR/ENF	5.4	110	E138A/E G140S Q148H	273.26	>892
M12	RAL/3TC/ETR/ENF	5.4	102	E138A/E G140S Q148H	ND	ND
Patient 2						
D0	RAL/TDF/3TC	4.6	16	V72I V201I	0.96	0.66
M1	RAL/TDF/3TC	2	53	ND	ND	ND
M3	RAL/TDF/3TC	<1.6	92	ND	ND	ND
M6	RAL/TDF/3TC	4.1	33	V72I G140S Q148H V201I	ND	ND
M9	RAL/TDF/3TC	4.3	49	V72I G140S Q148H V201I	215.34	>793
M12	RAL/TDF/3TC	4.08	28	V72I G140S Q148H V201I	449.72	>940
Patient 3						
D0	RAL/3TC/TDF/ETR	2.5	254	V201I	ND	0.87
M3	RAL/3TC/TDF/ETR	4.3	186	ND	ND	ND
M6	RAL/3TC/TDF/ETR	4.4	113	ND	ND	ND
M9	RAL/3TC/TDF/ETR	4.7	105	G140S Q148R K156N V201I	154.46	339.94
M12	RAL/3TC/TDF/ETR	4	75	G140S Q148R K156N V201I	326.48	847.95
Patient 4						
D0	RAL/3TC/ddI/ETR	4.7	7	V72I V201I T206S	0.69	1.03
M1	RAL/3TC/ddI/ETR	4.9	12	ND	ND	ND
M3	RAL/3TC/ddI/ETR	5	3	ND	ND	ND
M6	RAL/3TC/ddI/ETR	5	2	V72I L74M G140A Q148R V201I T206S	192.71	>646
M9	RAL/3TC/ddI/ETR	5.1	2	V72I L74M G140A Q148R V201I T206S	154.46	339.94
M12	RAL/3TC/ddI/ETR	5.1	3	V72I L74M G140A Q148R V201I T206S	326.48	847.95
Patient 5						
D0	RAL/FTC/ETR/DRV/r	4.4	117	V72I V151I/V T206S	1.57	2.12
M1	RAL/FTC/ETR/DRV/r	4.7	122	V72I G140S Q148H T206S	337.43	>793
M3	RAL/FTC/ETR/DRV/r	5.4	71	V72I G140S Q148H T206S	ND	ND
M6	RAL/FTC/ETR/DRV/r	6.2	45	V72I G140S Q148H T206S	482.58	>940
M9	RAL/FTC/ETR/DRV/r	6.4	17	V72I G140S Q148H T206S	647.83	>892
M12	RAL/FTC/ETR/DRV/r	6.4	17	V72I L74L/M G140S Q148H T206S	908.45	>940
Patient 6						
D0	RAL/ETR/DRV/r	6.7	16	V72I I203M	0.89	0.89
M1	RAL/ETR/DRV/r	5.8	20	V72I N155N/H I203M	1.07	2.1
M3	RAL/ETR/DRV/r	6.3	4	V72I L74L/M T97A/T Y143C I203M S203R	5.87	1.56
M6	RAL/ETR/DRV/r	5.6	10	V72I L74L/M T97A/T Y143C/Y I203M	ND	ND
M9	ETR/DRV/r	5.7	10	V72I I203M	0.98	0.83
M12	ETR/DRV/r	5.5	7	V72I I203M	0.76	1.43
Patient 7						
D0	RAL/ETR/DRV/r/ENF	6.4	36	V151I/V	0.76	0.94
M1	RAL/ETR/DRV/r/ENF	5.2	77	ND	ND	ND
M3	RAL/ETR/DRV/r/ENF	5.4	67	N155H	1.72	6.05
M6	RAL/ETR/DRV/r/ENF	4.8	36	T97A/T G163G/R Y143Y/C/H/R	28.52	6.23
M9	RAL/ETR/DRV/r/ENF	5.6	31	T97A/T G163G/R Y143Y/C/H/R	21.13	2.68
M12	RAL/ETR/DRV/r/ENF	5.5	12	L74L/M T97A/T G163G/R Y143Y/C/H/R V201I/V	ND	ND

Continued

Table 2. *Continued*

Time on RAL	ART	Viral load (log ₁₀ copies/mL)	CD4 (cells/mm ³)	Integrase mutations	RAL FC	EVG FC
Patient 8						
D0	RAL/TDF/FTC/ETR/DRV/r	5.6	22	V72I	0.91	0.60
M1	RAL/TDF/FTC/ETR/DRV/r	4.7	48	V72I	ND	ND
M3	RAL/TDF/FTC/ETR/DRV/r	4.5	88	V72I N155N/H	ND	ND
M6	RAL/TDF/FTC/ETR/DRV/r	5.6	103	V72I T97A/T Y143C/H/R/Y	51.95	3
M12	RAL/TDF/FTC/ETR/DRV/r	6.2	69	V72I T97A/T Y143C/H/R/Y	ND	9.57
Patient 9						
D0	RAL/ddI/TDF/ETR/DRV/r	5.4	13	V72I V201I	0.75	0.96
M1	RAL/ddI/TDF/ETR/DRV/r	2.9	218	ND	ND	ND
M3	RAL/ddI/TDF/ETR/DRV/r	2.2	190	S230N	ND	ND
M6	RAL/ddI/TDF/ETR/DRV/r	1.9	165	ND	ND	ND
M9	RAL/ddI/TDF/ETR/DRV/r	1.8	250	ND	ND	ND
M12	RAL/ddI/TDF/ETR/DRV/r	<1.6	372	ND	ND	ND
Patient 10						
D0	RAL/3TC/ETR/DRV/r	3.7	300	V72I	1.06	1.29
M1	RAL/3TC/ETR/DRV/r	<1.6	231	ND	ND	ND
M3	RAL/3TC/ETR/DRV/r	<1.6	240	ND	ND	ND
M6	RAL/3TC/ETR/DRV/r	1.95	258	ND	ND	ND
M9	RAL/3TC/ETR/DRV/r	1.83	256	ND	ND	ND
M12	RAL/3TC/ETR/DRV/r	<1.6	382	ND	ND	ND
Patient 11						
D0	RAL/FTC/TDF/DRV/r	5.76	124	V201I	2.18	1.54
M1	RAL/FTC/TDF/DRV/r	3.74	74	ND	ND	ND
M3	RAL/FTC/TDF/DRV/r	3.81	122	V201I	ND	ND
M6	RAL/FTC/TDF/DRV/r	2.82	104	V201I	ND	ND
M9	RAL/FTC/TDF/DRV/r	3.46	80	V201I	0.64	1.24
M12	RAL/FTC/TDF/DRV/r	5.57	55	V201I	1.39	2.28

Time on RAL, duration of raltegravir-based therapy (months); RAL FC, raltegravir sensitivity fold change; EVG FC, elvitegravir sensitivity fold change; 3TC, lamivudine; TDF, tenofovir diphosphofumarate; ETR, etravirine; ddI, didanosine; DRV/r, darunavir boosted with ritonavir; ENF, enfuvirtide; FTC, emtricitabine; ND, not determined.

Table 3. Comparisons of raltegravir C_{min} in the presence or absence of different patterns of HIV-1 integrase inhibitor resistance mutations

Time on RAL	RAL C_{min} (IQR) (ng/mL)			P^*
	Q148H/R (n=5)	155/Y143C/H/R (n=3)	WT or VR (n=43)	
M1	350 (300–350)	100 (50–150)	250 (200–350)	0.06
M3	200 (100–350)	75 (50–100)	300 (200–350)	0.03
M6	100 (100–150)	75 (50–75)	300 (200–350)	0.006

Time on RAL, duration of raltegravir-based therapy (months); WT or VR, failure with wild-type virus or virological response; IQR, interquartile range.

* P value determined using a Kruskal–Wallis test.

median elvitegravir FC was 847.95 (range: 339.94–640). The two samples with the N155H mutation had low raltegravir FCs (1.07 and 1.72) and low elvitegravir FCs (2.1 and 6.05). In samples with

the Y143C/H/R mutations, the median baseline raltegravir FC was 24.83 (range: 5.87–51.95) and the median baseline elvitegravir FC was 3 (range: 1.56–9.57). No phenotypic result could be obtained for the sample with the S230N mutation, because of low viral load.

Discussion

We characterized the genotypic evolution of HIV-1 integrase in patients with VF on raltegravir. These patients represented 11/51 (22%) antiretroviral-experienced patients who had received raltegravir together with an optimized background therapy. Undetectable viral loads were obtained in 78% of patients, similar to previously reported rates of virological response in clinical trials of raltegravir in antiretroviral-experienced patients.^{7,8} HIV-1 integrase resistance mutations were selected in 9/11 (81%) patients, with two main pathways. As previously reported,⁹ the most frequent pathway involved the Q148H/R mutations associated with the G140S/A mutations. The

Q148H/R mutations have been reported to be associated with high levels of phenotypic resistance to raltegravir.¹⁷ The G140S mutation has been recently shown to rescue the catalytic defect due to the Q148H mutation, enabling the recovery of the viral fitness.¹⁸ We confirmed the high level of phenotypic resistance to raltegravir in five patients with the Q148H/R pathway which was associated with high levels of plasma viral loads.

The second resistance pathway was detected in three patients and corresponded first to the selection of the N155H mutation, subsequently replaced by a pattern involving the Y143R/C/H and T97A mutations associated with L74M, G163R or S230R. The mutation N155H had been previously detected as an alternative pathway of resistance to raltegravir,⁹ whereas the Y143R/C/H mutations had been rarely found in patients failing on raltegravir.¹⁹ The replacement of N155H by Y143R/C/H suggested an advantage for replication of this latter pattern. A marked decrease in phenotypic sensitivity to raltegravir was observed for the isolates harbouring the Y143R/C/H mutations; in contrast, little phenotypic resistance to raltegravir was noted in the presence of N155H; however, the low FC could in this case be due to the presence of wild-type virus in the global population because N155N/H was detected in a population mixture by sequencing. We can thus speculate that the increase of resistance to raltegravir with the Y143R/C/H mutation might have led to the selection of these mutations and to the disappearance of N155H. Another explanation could be that the viral evolution was driven by the disadvantage in replicative capacity conferred by the N155H mutation.¹⁷ The emergence of different integrase mutation patterns in patients failing raltegravir could be influenced by various factors. Interestingly, the plasma raltegravir C_{min} was lower in patients with the selection of the N155H mutation followed by Y143C/H/R compared with patients with the Q148H/R mutation and with patients without emerging mutations or with virological response. This finding has to be taken together with the measurement of lower phenotypic FCs with N155H or Y143C/H/R than with Q148H/R. It suggests that in the presence of low raltegravir plasma levels the 155/143 pathways could be advantaged whereas higher but still suboptimal raltegravir plasma levels could promote the high-level resistance pathway encoded by Q148H/R+G140S/A. Even if we have to be cautious due to the small numbers of resistant isolates included in this comparison, we think that this finding is of clinical importance since it suggests that a suboptimal adherence to raltegravir-based therapy could have a high cost with regard to the selection of resistant viruses. The two latter patients failed therapy without emerging integrase mutations although raltegravir exposure appeared sufficient; this could be due to ongoing viral replication in viral reservoirs where inhibitory raltegravir concentrations could not be reached.

Another factor which could explain the selection of different raltegravir resistance pathways is the baseline variability of the integrase gene. However, we could not find any association in our patients between the presence of polymorphic substitutions at baseline and the subsequent selection of the different patterns. Further studies including a larger number of patients will be needed to determine precisely both the pharmacokinetic and virological factors influencing the raltegravir resistance pathways.

Our study also provides interesting data concerning the sensitivity to elvitegravir of raltegravir-resistant isolates. Differences in susceptibility to elvitegravir were noted according to the different patterns of raltegravir resistance mutations. High-level resistance to elvitegravir was shown in the presence of the Q148H/R pathway. N155H was associated with low-level resistance to elvitegravir in our study, which could be due to the presence of a mixture of population N155N/H. This, however, confirms data obtained with site-directed mutants¹² showing a large reduction in susceptibility to elvitegravir in the presence of Q148R but substantially lower levels of resistance to elvitegravir with N155H. The isolates with the Y143C/H/R mutations showed little phenotypic resistance to elvitegravir in our patients. This seems to be in accordance with recent findings²⁰ showing frequent selection of T66I/A/K, E92Q, Q148H/R/K and N155H but not of Q143C/H/R in patients with VF on elvitegravir.

Although different levels of cross-resistance to elvitegravir are encoded by the different raltegravir resistance mutations, future generations of INIs will probably need to act through a different mechanism and/or interact with other portions of the integrase enzyme in order to benefit raltegravir-failing patients. It could, however, be of importance to prevent the continuous selection of new raltegravir resistance mutations or pathways in patients with VF on raltegravir in order to preserve future options with next-generation INIs. We report in our study the selective withdrawal of raltegravir in one patient with the mutation Y143C, with no significant increase in HIV-1 replication. Similar findings were observed in a recent study on four individuals with either N155H or Q148H, selectively stopping raltegravir,²¹ suggesting limited residual activity of raltegravir in these patients. However, we should be cautious in interpreting data on limited numbers of patients. The disappearance of the Y143C mutation from the plasma sample following the withdrawal of raltegravir is likely to indicate a disadvantage in viral fitness conferred by this mutation.

In conclusion, we characterized the dynamics of the different integrase genotypic patterns in patients failing on raltegravir-containing regimens. Pharmacokinetic and virological parameters should be further assessed in larger studies in order to confirm their role as determinants of the genotypic pathways of resistance to raltegravir. A precise understanding of the *in vivo* and *in vitro* mechanisms of resistance to raltegravir will be of interest to preserve future options with next-generation INIs.

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Transparency declarations

We have no conflicts of interest to declare.

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